Table S1. Characteristics of children included in studies comparing progesterone versus placebo

Name first author and year of original trial	Name first author and year of article with long-term outcomes	Singleton or multiple pregnancies	Maternal treatment	Number of children assessed	Age of children at assessment in months, mean (SD)	Gestational age at delivery in weeks, mean(SD), or median (IQR)
Meis 2003	**Northen 2007**	Singleton	Progesterone	194	48 months	Delivery before 37
		pregnancies	Placebo	84		weeks gestation [§]
						Prog 36.3%
						Plac 54.9%
Norman 2009	**McNamara 2015**	Multiple	Progesterone	140	55 (10) (n=69)	36 (3) (n=74)
STOPPIT	STOPPIT	pregnancies	Placebo	184	55 (11) (n=91)	36 (2) (n=91)
Rode 2011	**Rode 2011**	Multiple		6months/18months	6 & 18 months	
PREDICT	PREDICT	pregnancies	Progesterone	514 / 501		36.0 (2.8)*
			Placebo	536 / 490		35.9 (2.7)*
	Klein 2011	Multiple		6months/18months	6 & 18 months	
	PREDICT	pregnancies	Total	112 / 102		36.0 (2.8)*
						35.9 (2.7)*
	Vedel 2016	Multiple	Progesterone	225	56.9 (6.0)	37.28 (35.14 - 38.0)
	PREDICT	pregnancies	Placebo	212	57.1 (6.1)	37.0 (35.14 - 38.0)
Van Os 2015	**Cuijpers**					
TripleP	***unpublished***	Singleton	Progesterone	29	25 (23-27) [§]	38.85 (37.14 - 40.28)
	TripleP	pregnancies	Placebo	30	25 (23-27)	38.71 (37.85-40.14)
Norman 2018	**Norman 2018**	Singleton	Progesterone	430	24.7 (3.9)	36.71 (4.1)*
OPPTIMUM	OPPTIMUM	pregnancies	Placebo	439	24.8 (4.08)	36.85 (4.2)*

[§] No report of a mean/median gestational age at delivery, only dichotomous outcome; preterm delivery before 37 weeks of gestation, before 35 weeks of gestation and before 32 weeks of gestation

*Gestational age at delivery from entire cohort (gestational age at delivery follow-up participants not available)

Table S2. Overview of reported statistical analyses of included follow-up studies

Name first author and year of article with long-term outcomes	**Statistical analyses of follow-up outcomes, as reported in the methods paragraph**	**Reported effect measures** Risk Ratio (RR), Odds ratio (OR), Mean Difference (MD), Effect Size (ES) or p-value
Northen 2007	Multivariable analysis controlling for centre was performed by logistic regression.	p-value
McNamara 2015 STOPPIT	Outcomes were analysed using generalised linear models (linear of logistic regression) where appropriate, adjusted for the clustering at the level of the mother.	OR, MD, ES
Rode 2011 PREDICT	Binary outcomes were analysed by logistic regression models and continuous outcomes by linear regression models. The correlation within pairs of twins was accounted for by considering robust estimates of variable based on the method of generalized estimating equations (GEE). GEE models for mean ASQ scores were adjusted for time between estimated date of delivery and the date on which the questionnaire was completed by the parents.	OR, p-value
Klein 2011 PREDICT	Categorical variables were analysed using chi-square of Fisher's exact test and continuous variables using unpaired t-test, Kruskal-Wallis and Mann-Whitney U. Binary outcomes were analysed by logistic regression models.	OR, p-value
Vedel 2016 PREDICT	For outcomes of the children, the correlation within pairs of twins was accounted for by considering robust estimates of variance based on the method of generalized estimating equations. All analyses were unadjusted.	OR, p-value
Cuijpers *2020* TripleP	A Directed Acyclic Graph was used to visualise potential confounders. This graph demonstrated no potential confounders and therefore no correction was applied. The composite outcome with imputed data was analysed with a generalized linear model.	RR, OR, MD
Norman 2018 OPPTIMUM	Mixed effects logistic regression (for continuous variables linear regression) models including treatment allocation and previous pregnancy (≥14 weeks) as fixed effects, with study centre as random effect. P-values were reported without adjustment for multiple comparisons, then adjusted using a Bonferroni-Holm procedure.	OR, MD

Table S3. Quality in Prognostic Studies appraisal (QUIPS)

Quality in prognostic studies (QUIPS)							
	Northen 2007	**McNamara 2015**	**Rode 2011**	**Klein 2011**	**Vedel 2016**	**Cuijpers 2020**	**Norman 2018**
Study participation							
Adequate participation	✓	✓	✓	✓	x	✓	✓
Source of population	✓	✓	✓	✓	✓	✓	✓
Baseline study sample	✓	✓	✓	✓	✓	✓	✓
Sampling frame & recruitment	✓	✓	✓	✓	✓	✓	✓
Period & place of recruitment	✓	✓	✓	✓	✓	✓	✓
Inclusion & Exclusion criteria	✓	✓	✓	✓	✓	✓	✓
Assessment of bias	Low bias	Low bias	Low bias	Low bias	Moderate bias	Low bias	Low bias
Study Attrition							
Response rate	✓	/	✓	✓	x	✓	✓
Information about participants who dropped out	✓	✓	x	x	x	✓	✓
Reasons for loss-to-follow-up	✓	x	x	x	x	✓	✓
Description of participants lost to follow-up	/	✓	x	x	✓	✓	✓
Differences between participants	✓	x	x	x	x	✓	✓
Assessment of bias	Low bias	High bias	High bias	High bias	High bias	Low bias	Low bias
Prognostic factor measurement							
Definition of PF	✓	✓	✓	✓	✓	✓	✓
Valid and reliable measurement of PF	✓	✓	✓	✓	x	✓	✓
Continuous variables/cut points	✓	✓	✓	✓	✓	✓	✓
Consistency of method and setting of measurement	✓	✓	✓	✓	✓	✓	✓
Proportion of data available for analysis	✓	✓	✓	✓	✓	✓	✓
Method used for missing data	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Assessment of bias	Low bias	Low bias	Low bias	Low bias	High bias	Low bias	Low bias
Outcome measurement							
Definition of the outcome	✓	x	✓	✓	✓	✓	✓
	Northen 2007	**McNamara 2015**	**Rode 2011**	**Klein 2011**	**Vedel 2016**	**Cuijpers 2020**	**Norman 2018**

Valid and reliable measurement of outcome	/	✓	✓	✓	/	✓	✓
Consistency of method and setting of measurement	✓	✓	✓	✓	✓	✓	✓
Assessment of bias	Moderate bias	Low bias	Low bias	Low bias	High bias	Low bias	Low bias
Study Confounding							
Not applicable							
Statistical analysis and reporting							
Sufficient presentation of data to assess adequacy of analytic strategy	✓	✓	/	/	✓	✓	✓
Adequate strategy for model building	✓	✓	/	/	✓	✓	✓
Adequate statistical model	✓	✓	/	/	✓	✓	✓
No selective reporting	✓	✓	/	/	✓	✓	✓
Assessment of bias	Low bias	Low bias	High bias	High bias	Low bias	Low bias	Low bias

✓ fulfilled
/ partially fulfilled
X not fulfilled
? unknown
n/a not applicable

Table S4. Summary of offspring general health outcomes

General health – components of disability and/or diagnosis					
	Age at follow-up	**Number of children progesterone/ placebo**	**Progesterone n(%)**	**Placebo n(%)**	**Risk Ratio (RR) (95%CI), Odds ratio (OR) (95% CI), Mean Difference (MD) (95% CI), Effect Size (ES) (95% CI)or p-value**
Motor function*					
Norman 2018	2 years	461/456	4 (1)	4 (1)	OR 0.99 (0.25 - 3.98)
Northen 2007	4 years	192/82	2 (1)	1 (1.2)	p-value 1.00
Cognitive function*					
Norman 2018	2 years	461/452	19 (4)	18 (4)	OR 1.03 (0.58 - 1.84)
Northen 2007	4 years	192/82	14 (7.2)	7 (8.3)	p-value 0.75
Hearing					
Norman 2018	2 years	466/465	1 (<1)	2 (<1)	OR 0.56 (0.33 - 0.94)
McNamara 2015	Neonatal hearing screening	302/301	41 (14)	31 (10)	ES 1.38 (0.81 - 2.35)
Speech and Language*					
Norman 2018	2 years	445/446	18 (4)	14 (3)	OR 1.32 (0.72 - 2.43)
Northen 2007	4 years	192/82	9 (4.7)	7 (8.5)	p-value 0.26
Vision					
Norman 2018	2 years	447/466	0	4(1)	n/a
Vedel 2016	up to 8 years	492/497	11 (2.2)	11 (2.2)	OR 1.03 (0.41 - 2.55)
McNamara 2015	Preschool vision screening	250/263	70(28)	71(27)	ES 1.04 (0.67 - 1.64)
Respiratory problems / Asthma diagnosis*					
Norman 2018	2 years	413/434	7(2)	3(1)	3.03 (1.56 - 5.88)
Vedel 2016	up to 8 years	492/497	34 (6.9)	27 (5.4)	OR 1.29 (0.72 - 2.32)
Northen 2007	4 years	192/82	39 (20.3)	20 (24.4)	p-value 0.45
Gastrointestinal					
Norman 2018	2 years	412/432	9(2)	4(1)	OR 2.67 (1.37 - 5.20)
Vedel 2016	up to 8 years	492/497	29 (5.9)	28 (5.6)	OR 1.04 (0.58 - 1.87)
Northen 2007	4 years	192/82	5(2.6)	1(1.2)	p-value 0.30
Renal					
Norman 2018	2 years	414/434	3(1)	1(<1)	OR 3.65 (1.96 - 6.82)
Vedel 2016	up to 8 years	492/497	5 (1.0)	1 (0.2)	OR 5.09 (0.59 - 43.68)
Any infection					
Vedel 2016	up to 8 years	492/497	177 (36.0)	164 (33.0)	OR 1.14 (0.84 - 1.54)
Neurological					

	Age at follow-up	Number of children progesterone/ placebo	Progesterone n(%)	Placebo n(%)	Risk Ratio (RR) (95%CI), Odds ratio (OR) (95% CI), Mean Difference (MD) (95% CI), Effect Size (ES) (95% CI)or p-value
Vedel 2016	up to 8 years	492/497	40 (8.1)	41 (8.3)	OR 0.98 (0.59 - 1.61)
Northen 2007	4 years	192/82	0 (0)	1 (1.2)	p-value 1.0
Cuijpers 2020	2 years	25/25	2 (8.0)	1 (4.0)	RR 2.00 (0.19 - 20.67)
Cardiac					
Vedel 2016	up to 8 years	492/497	24 (4.9)	15 (3.0)	OR 1.66 (0.81 - 3.37)
Dermatological					
Vedel 2016	up to 8 years	492/497	15 (3.1)	22 (4.4)	OR 0.67 (0.32 - 1.41)
Northen 2007	4 years	192/82	35 (18.2)	12 (14.6)	p-value 0.47
Skeletal					
Vedel 2016	up to 8 years	492/497	13 (2.6)	10 (2.0)	OR 1.34 (0.54 - 3.36)
Growth – Height					
McNamara 2015	Screening 4.5-5.5 years Mean cm	165/189	111 (5)	112 (5)	MD -0.6 (-2.1-0.8)
McNamara 2015	Screening 4.5-5.5 years Percentile	165/189	42.1 (28)	48.5 (28.8)	MD -6.4 (-12.3- -0.4)
Norman 2018	2 years Mean	347/369	87.4 (7.9)	87.2 (10.7)	Not reported
Northen 2007	48 months Percentile	189/81	54 (29)	57 (29)	p-value 0.53
Growth – Weight					
McNamara 2015	Screening 4.5-5.5 years of age Mean kg	165/189	19.4 (3.2	19.7 (2.8)	MD 0.2 (-1.0-1.4)
McNamara 2015	Screening 4.5-5.5 years of age Percentile	165/189	43.5 (29.7)	48.6 (29.2)	MD -5.2 (-11.3-1.0)
Norman 2018	2 years Mean kg	332/355	13.4 (2.7)	13.2 (2.6)	Not reported
Northen 2007	48 months Percentile	189/81	55 (30)	57 (30)	p-value 0.66
BMI					
McNamara 2015	Screening 4.5-5.5 years of age	165/189	15.6 (2)	15.6 (1.5)	MD 0.2 (-0.3-0.4)
Admission to hospital during follow-up					
	Age at follow-up	**Number of children progesterone/**	**Progesterone n(%)**	**Placebo n(%)**	**Risk Ratio (RR) (95%CI), Odds ratio (OR) (95% CI), Mean Difference (MD) (95% CI),**

		placebo			Effect Size (ES) (95% CI)or p-value
Norman 2018	2 years	416/434	48(12)	51(12)	OR 0.98 (0.65 - 1.47)
Vedel 2016	up to 8 years	492/497	246 (50.0)	264 (53.1)	OR 0.88 (0.66 - 1.17)
McNamara 2015	up to 5 until 9 years	386/395	159 (41)	165 (42)	ES 0.97 (0.71 - 1.33)
Northen 2007	4 years - any operations	192/82	23(12)	17(20.7)	p-value 0.06
Cuijpers 2020	2 years	29/25	2 (6.9)	2 (8.0)	RR 0.86 (0.13 - 5.68)
Multi-attribute Health Status					
McNamara 2015	5 years - HUI mark II	147/184	0.96 (0.1)	0.96 (0.09)	MD 0.00 (-0.03 - 0.02)
	5 years - HUI mark III	147/184	0.96 (0.12)	0.97 (0.07)	MD -0.01 (-0.03 - 0.02)
Congenital malformation / chromosomal anomalies					
Rode 2011	Neonatal	663/677	25(3.8)	27(4.0)	OR 1.0 (0.5 - 1.7)
McNamara 2015	up to 5 until 9 years	386/395	17 (4)	17 (4)	ES 1.04 (0.49 - 1.21)
Cuijpers 2020	up to 2 years	26/26	7 (26.9)	3 (11.5)	RR 2.33 (0.68 - 8.05)
Genital abnormality					
Rode 2011	Neonatal	663/677	6 (0.9)	8 (1.1)	
Vedel 2016	up to 8 years	492/497	23 (4.7)	22 (4.4)	OR 1.06 (0.55 - 2.03)
Cuijpers 2020	up to 2 years of age	25/26	3(12.0)	3(11.5)	RR 0.96 (0.21-4.3)

*Not assessed with an internationally validated questionnaire or physical test

Table S5. Summary of offspring behavioural outcomes

Behaviour						
	Age at follow-up	**Score**	**Number of children progesterone/ placebo**	**Progesterone mean (SD) or n(%)**	**Placebo mean (SD) or n(%)**	**Odds ratio (95% CI) or p-value**
SDQ total difficulties						
Norman 2018	2 years	Mean	295 / 302	10.2 (4.9)	9.8 (4.9)	OR 1.23 (0.85 - 1.78)
CBCL						
Cuijpers 2020	2 years	Cut-off†	27/27	1 (3.7)	4 (14.8)	OR 0.22 (0.02 - 2.12)

† Defined as a score in the clinical range (>97th percentile) in any of the subscales or a score ≥90th percentile in one of the two broad dimensions or total problem score was denied as abnormal